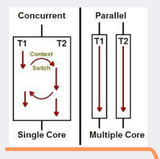
Java Reactive

0. Basic concepts

Parallelism vs Concurrency



Parallelism is about doing lot of things at once.

- Tools
  - Threads
  - Thread Pool
    - ExecuterService
    - ForkJoinPool
    - Custom Thread Pools (eg: Web Servers)
    - Requires >1 cpu cores.

Concurrency is about dealing with lot of things at once

- Shared resource is to be accessed/updated
- Multiple tasks need to coordinate
- Tools
  - Locks / synchronized
  - Atomic classes
  - Concurrent data structures
    - ConcurrentHashMap
    - BlockingQueue
  - CompletableFuture
  - CountDownLatch / Phaser / CyclicBarrier / Semaphore, etc

Funcional Programming

- It's a paradigm that treats computation as the evaluation of mathematical functions and avoids changing-state and mutable data.
- Combine Reactive Programming and Functional programming to write code in a faster, clearer and more understandable way.

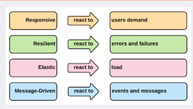
Streams API

- Streams let you process data in a declarative way
- Pipelining: Chain of various stream operations.

The main goal

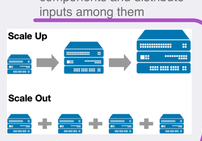
- To maximize the use rate of modern multicore CPUs and, more precisely, of the threads computing for their use.
- Code is more waiting than working

Reactive Manifesto



https://www.reactivemanifesto.org/

- Responsive
  - The system responds in a timely manner if possible
  - This is key for usability and utility
  - Consistent response times
- Resilience
  - The system stays responsive when facing a failures
  - Performance, endurance and security are all facets of resiliency
- Elastic
  - The system stays responsive under varying workload
  - Design with no contention points or bottlenecks
  - Ability to share or replicate components and distribute inputs among them



Scaling up or out

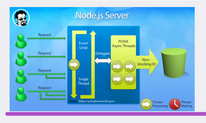
- Message-Driven
  - Reactive applications rely on asynchronous message-passing to establish a boundary between components
  - Ensure loose coupling, isolation and location transparency.
  - Delegate failures as messages
  - Non-blocking communication allows recipients to only consume resources while active

3. Features

Nonblocking

- Vs Blocking
  - Single Item: T
  - Many Items: Stream<T>
- Project Reactor
  - Single Item: Mono <T>
  - Many Items: Flux<T>
  - https://projectreactor.io/

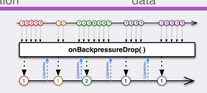
- Blocking



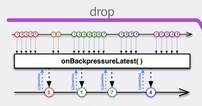
- Other Frameworks

Back pressure (Throttling)

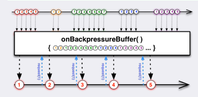
- Problem: Source of information overflows the consumer with data
- Solution
- Mechanism



drop



latest



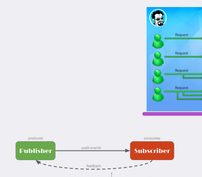
buffer

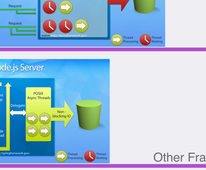
- Provides resilience

Cold vs Hot Sources of information

- A cold source of information start running upon subscription
- A hot source of information is always emitting data, even before a subscription is active

Cold Publisher





Hot Publisher

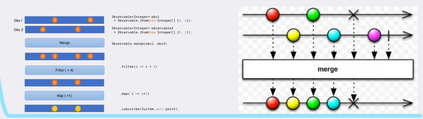
Otros

- Multiple Subscribers
- Less latency
- More tolerance of failure
- Threads
  - Using very few threads
  - Low-level threading
  - Thread safety

1. Definitions

Reactive Programming

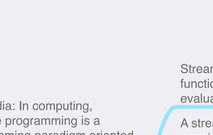
- Wikipedia: In computing, reactive programming is a programming paradigm oriented around data flows and the propagation of change
  - Stream is an internal iterator functional composition and Lazy evaluation.
  - A stream is a sequence of ongoing events ordered in time
  - Declarative, Event based programming
  - Decompose problem into multiple, asynchronous discrete steps.
- Example simple: Spreadsheet
  - Define a formula in a cell.
  - How the change is propagated when you change the value of one cell.
- Reactive Streams
  - Standard for asynchronous stream processing with Non-blocking and Back pressure
- It's more than just async and non blocking execution
  - Advanced time scheduling
  - Flexible Scheduling
  - Back pressure control
  - Resilience on errors



2. Java Streams vs Reactive Streams

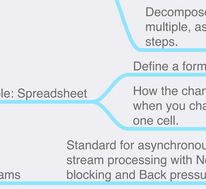
Pull vs Push Data

- With the imperative model we pull and process things synchronously using iterators.
- With reactive programming we take that and create push model.



Data only vs 3 Channel

- Data
- Error
- Complete



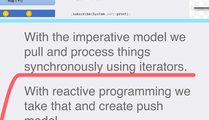
Exceptions vs Circuit breaks

- Error is just another form of data in reactive programming.
  - First Class Citizen
- Exceptions are imperative style ideas.

Secuential vs Parallel

Single Pipeline vs Multiple subscribers

Synch vs Async



When to use

- Java Streams
  - Collection Processing
  - Batch computational jobs
  - Queries on already available data
    - Find the transaction with the highest value
    - Find the green apples over 150 grams.
- Observables
  - On demand computation
  - Updating calculations based upon a series o events
  - I/O workloads
  - Eg:
    - Aggregate saved video positions stored on different micro services.
    - Monitor stock market price ticks until the price is below 100
    - Aggregating temperature sensor information to control heating

4. Api

Java 9 Reactive Stream Api

- java.util.concurrent.Flow
  - Publisher
    - A potentially unbounded source of events.
  - Subscriber
    - Receives events.
  - Subscription
    - Control the processing + back pressure of the events.
  - Processors
    - Both a Publisher and Subscriber.
- Used by Libraries
- JDK Standardization of the Reactive Stream Protocol.
- Enables greater adoption by having it in the core JDK.
- https://github.com/reactive-streams/reactive-streams-jvm
- Some rules
  - If a publisher fails it must signal an on Error
  - If a Publisher terminates successfully (finite stream) it must signal an on Complete
  - If a Subscription is cancelled its Subscriber MUST eventually stop being signaled
  - Publisher.subscribe MAY be called as many times as wanted but MUST be with a different Subscriber each time
  - A subscriber MUST signal demand via Subscription.request(long n) to receive onNext signals

RxJava

- Used directly by Programmers
- Why RxJava
  - Callbacks were too low level an API
  - Completable Future only deals with a single value.
    - What if you are trying process a series of values on the fly.
    - 3 channel vs 2 channel
    - Reactive Stream vs CompletableFuture/Promises
      - CompletableFuture<List<V>>
        - Wait for all elements in the list to be available.
      - Observable<V>
        - Start processing one by one on arrival
  - Actors require you to buy into a whole actor framework and are also a lower lever API.

Alternative Implementations

- Frameworks/Libs
  - Project Reactor - Spring
  - AkkaStreams - Akka
  - Vert.x
  - RxNetty
- Reactive socket.io
  - Web socket protocol with back pressure.
  - https://rsocket.io/

Reactive Systems

- An architectural style that allows multiple applications to work together.
- https://projectreactor.io/
- https://spring.io/reactive
  - Spring WebFlux
    - Mono
    - Flux